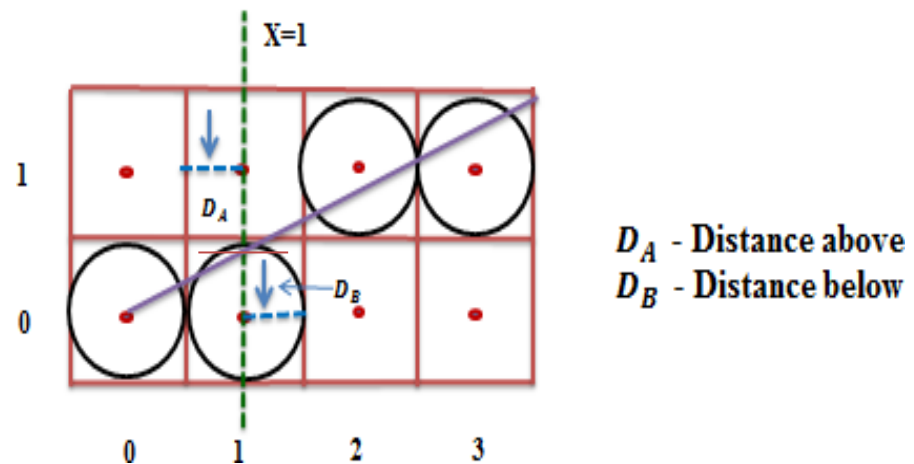
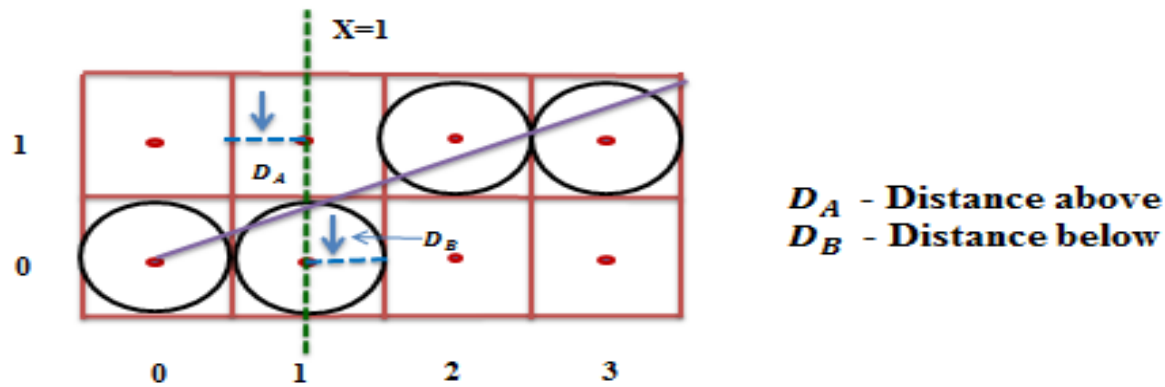


Bresenham's Line Algorithm

The *basic principle of Bresenham's line algorithm* is to **select the optimum raster locations** to **represent a straight line**.

- To accomplish this algorithm always **increments** either **x** or **y** by **one unit** **depending** on the **slope of line**.
- The increment in the other variable is determined by **examining the distance between** the **actual line location** and the **nearest pixel**.
- This **distance** is called **decision variable** or **the error**. This is illustrated in the Figure below.
- As shown above figure, the line **does not pass through all raster points (pixels)**. It **passes through raster point (0, 0)** and subsequently crosses three pixels.





It is seen that the intercept of line with the **line $x = 1$** is closer to the **line $y = 0$** , i.e. pixel (1, 0) than to the **line $y = 1$** i.e. pixel (1, 1).

- Hence, in this case, the **raster point** at (1, 0) **better represents the path of the line** than that at (1, 1).

The intercept of the line with the **line $x = 2$** is close to the **line $y = 1$** , i.e. pixel (2, 1) than to the **line $y = 0$** , i.e.: pixel (2, 0).

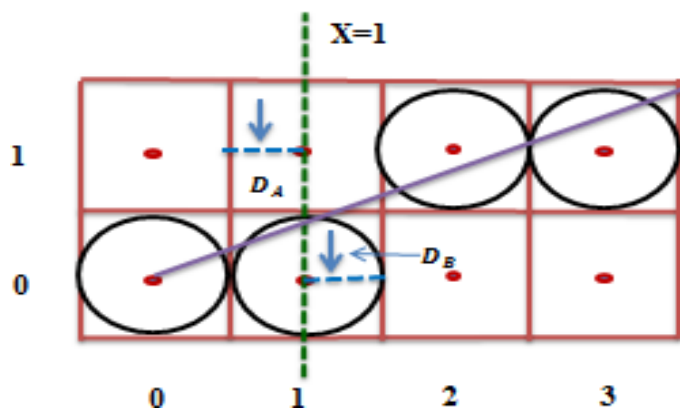
- Hence, the **raster point** at (2, 1) **better represents the path of the line**, as shown in the above figure.

In mathematical terms error or decision variable is defined as

$$e = D_B - D_A \quad \text{OR} \quad e = D_A - D_B$$

Let us define $e = D_B - D_A$. Now

- if ($e > 0$) then it implies that ($D_B > D_A$) i.e., the *pixel above the line* is *closer to the true line*.
- if ($e < 0$) then it implies that ($D_B < D_A$) i.e., the *pixel below the line* is *closer to the true line*.
- Thus by *checking* only the *sign of error* term it is possible to *determine the better pixel to represent the line path*.



D_A - Distance above
 D_B - Distance below

The error is initially set as

$$e = 2 \Delta y - \Delta x$$

Where

$$\Delta y = y_2 - y_1, \text{ and}$$

$$\Delta x = x_2 - x_1$$

Then according to value of e following actions are taken.

While ($e \geq 0$)

{

$$y = y + 1$$

$$e = e - 2 * \Delta x$$

}

$$x = x + 1$$

$$e = e + 2 * \Delta y$$

- When ($e \geq 0$), error is initialized with $e = e - 2 * \Delta x$. This is continued until **error is negative**. In each iteration y is *increment by 1*.
- When ($e < 0$), error is initialized to $e = e + 2 * \Delta y$. In both the cases x is *incremented by 1*.

Bresenham's Line Algorithm

1. Read the line end points (x_1, y_1) and (x_2, y_2) such that they are not equal.
[If equal, then plot that point and exit]
2. $\Delta x = |x_2 - x_1|$ and $\Delta y = |y_2 - y_1|$
3. $x = x_1$ [Initialize starting point]
 $y = y_1$
4. $e = 2 * \Delta y - \Delta x$ [Initialize value of decision variable or error to compensate for non zero intercepts]
5. $i = 1$ [Initialize counter]
6. Plot (x, y)
7. While $(e \geq 0)$
 - {
 - $y = y + 1$
 - $e = e - 2 * \Delta x$
 - }
 - $x = x + 1$
 - $e = e + 2 * \Delta y$
8. $i = i + 1$
9. If $(i \leq \Delta x)$ then go to step 6
10. Stop

Example:- Consider the line from (5, 5) to (13, 9), use the Bresenham's algorithm to rasterize the line

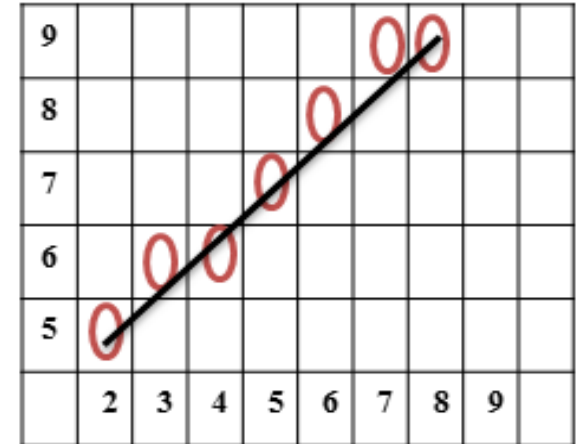
1. $\Delta x = |x_2 - x_1|$
2. $\Delta y = |y_2 - y_1|$
3. $x = x_1$
 $y = y_1$
4. $e = 2 * \Delta y - \Delta x$
5. $i = 1$
6. Plot (x, y)
7. While ($e \geq 0$)
 - {
 - $y = y + 1$
 - $e = e - 2 * \Delta x$
 - }
 - $x = x + 1$
 - $e = e + 2 * \Delta y$
8. $i = i + 1$
9. If ($i \leq \Delta x$) then
go to step 6
10. Stop

$\Delta x = |9 - 2| = 7$
 $\Delta y = |10 - 5| = 5$
 $x = 2$
 $y = 5$
 $e = 2 * 5 - 7 = 10 - 7 = 3$
 $i = 1$

6: plot (2, 5)
 while ($e \geq 0$)
 {
 $y = 5 + 1 = 6$
 $e = 3 - 2 * 7 = -11$
 }
 $x = 2 + 1 = 3$
 $e = -11 + 2 * 5 = -1$
 $i = i + 1 = 2$
 if ($2 \leq 7$) then
 goto step 6

6: plot (3, 6)
 while ($e \geq 0$)
 {
 $y = 6$
 $e = -1$
 }
 $x = 3 + 1 = 4$
 $e = -1 + 2 * 5 = 9$
 $i = 2 + 1 = 3$
 if ($3 \leq 7$) then
 goto step 6

6: plot (4, 6)
 $y = 6 + 1 = 7$
 $e = 9 - 2 * 7 = -5$
 $x = 4 + 1 = 5$
 $e = -5 + 2 * 5 = 5$
 $i = 3 + 1 = 4$
 if ($4 \leq 7$)
 goto step 6



Example:- Consider the line from (5, 5) to (13, 9), use the Bresenham's algorithm to rasterize the line.

Solution: Evaluating steps 1 through 4 in the Bresenham's algorithm we have,

$$\Delta x = |13-5|=8$$

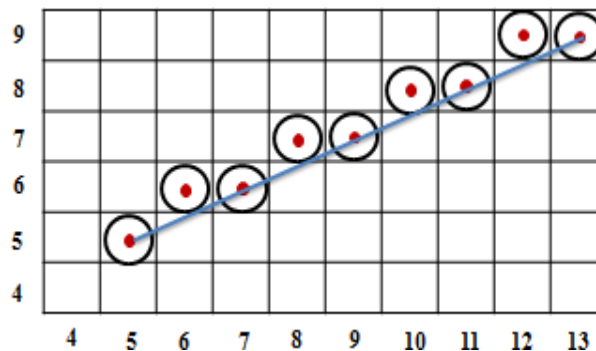
$$\Delta y = |9-5|=4$$

$$x=5, y=5.$$

$$e = 2 * \Delta y - \Delta x = 2 * 4 - 8 = 0$$

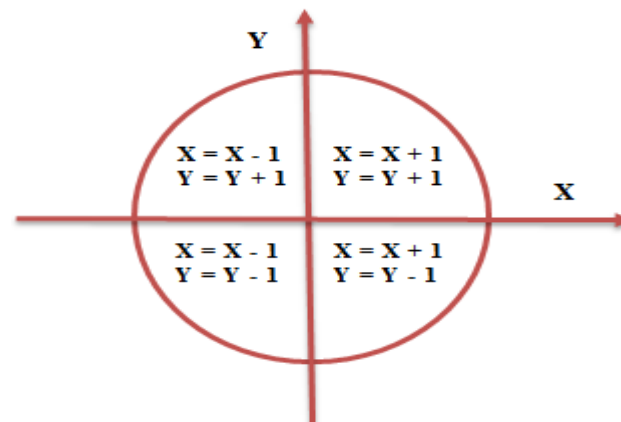
Tabulating the results of each iteration in the step 5 through 10

The results are plotted as shown in figure below



i	plot	x	y	e
		5	5	0
1	(5,5)	6	6	-16
				-8
2	(6,6)	7	6	0
3	(7,6)	8	7	-16
				-8
4	(8,7)	9	7	0
5	(9,7)	10	8	-16
				-8
6	(10,8)	11	8	0
7	(11,8)	12	9	-16
				-8
8	(12,9)	13	9	0
9	(13,9)	14	10	-16
				-8

- The Bresenham's algorithm *only works* for the *first octant*.
- The generalized Bresenham's algorithm *requires modification* for lines lying in the *other octants*.
- Such algorithm can be *easily developed* by *considering* the *quadrant* in which the *line lies* and *its slope*.
- When the *absolute magnitude* of the *slope of the line* is *greater than 1*, *y* is *incremented by one*.
- *Bresenham's decision variable* or *error* is used to *determine* when to *increment x*.
- The *x* and *y incremental values* depend on the *quadrant* in which the *line exists*. This is illustrated in below figure.



Conditions for generalized Bresenham's algorithm

Generalized Bresenham's Algorithm

1. Read the line end points (x_1, y_1) and (x_2, y_2) such that they are not equal.

[If equal then plot that point and exit]

2. $\Delta x = |x_2 - x_1|$ and $\Delta y = |y_2 - y_1|$

3. $x = x_1$ [Initialize starting point]
 $y = y_1$

4. $S_1 = \text{sign}(x_2 - x_1)$

$S_2 = \text{sign}(y_2 - y_1)$

5. If $\Delta y > \Delta x$ then

begin

$T = \Delta x : \Delta x = \Delta y : \Delta y = T : \text{Interchange} = 1$

end

else

$\text{Interchange} = 0$

End If

6. $e = 2 * \Delta y - \Delta x$ [Initialize value of decision variable or error to compensate for non zero intercepts]

7. $i = 1$ [initialize counter]

8. Plot (x, y)

9. While $(e \geq 0)$

begin

if ($\text{Interchange} = 1$) then $x = x + s_1$

else $y = y + s_2$

end If

$e = e - 2 * \Delta x$

end while

10. If ($\text{Interchange} = 1$), then $y = y + s_2$

else $x = x + s_1$

end If

$e = e + 2 * \Delta y$

11. $i = i + 1$

12. If ($i \leq \Delta x$) then go to step 8

13. Stop

Example : Draw the line from (0,0) to (-8,-4) using General Bresenham algorithm

$$\Delta x = | -8 - 0 | = 8$$

$$\Delta y = | -4 - 0 | = 4$$

$$x = x_1 \quad x_1 = 0$$

$$y = y_1 \quad y_1 = 0$$

$$S_1 = \text{sign}(x_2 - x_1) \\ = \text{sign}(-8 - 0) = \text{sign}(-8)$$

$$S_2 = \text{sign}(y_2 - y_1) \\ = \text{sign}(-4 - 0) = \text{sign}(-4)$$

$$x = 0 \quad y = 0$$

$$\Delta x = 8 \quad \Delta y = 4$$

$$S_1 = -1 \quad S_2 = -1$$

Because $\Delta x > \Delta y$ then

Interchange=0 :

$$e = 2 * \Delta y - \Delta x$$

$$e = 2 * 4 - 8$$

$$e = 0$$

i	plot	e	x	y
		0	0	0
1	(0,0)			
		-16	0	-1
		-8	-1	-1
2	(-1,-1)			
		0	-2	-1
3	(-2,-1)			
		-16	-2	-2
		-8	-3	-2
4	(-3,-2)			
		0	-4	-2
5	(-4,-2)			
		-16	-4	-3
		-8	-5	
6	(-5,-3)			
		0	-6	-3
7	(-6,-3)			
		-16	-6	-4
		-8	-7	-4
8	(-7,-4)			
		0	-8	-4